

Weekly Report: [W07]

General Information

- **Group ID:** 52
- **Group Name:** Group 52
- **Class:** 25A01
- **Project Name:** Super Mario Game
- **Date range:** 2026-07-19 – 2026-07-25
- **Github Repository:** <https://github.com/ndmhuy/SuperMarioGame.git>

Tasks Completed This Week

25125083 - Nguyễn Đình Minh Huy (Member A - Engine, Physics & Player/Item Entities)

Branch: `dev` (Integrated from `A/feature/save-load` & `A/feature/level-editor`)

- **Task 1: Feature Branches Integration & Code Review Fixes:** Merged task branches `A/feature/save-load` and `A/feature/level-editor` into `dev`. Addressed code review feedback for PR #5 & PR #6: fixed `EventBus` subscriber listener sentinel invalidation via `static_cast<size_t>(-1)`, replaced letter-based debug keys with `Num1` - `Num9`, and encapsulated keybindings accessors in `Game`.
- **Task 2: Achievement Toast Notification Queue Cleanup:** Resolved stale toast rendering bug on slot reloads inside `AchievementManager` by clearing the `m_activeToasts` queue in `setUnlockedAchievements()`, and aligned achievement string keys with `SPEC.md` schemas (`100_coins`, `beat_bowser`).
- **Task 3: Consolidated Unit Test Runner (`verify_all`):** Created a unified test harness `verify_all.cpp` combining Map Editor command undo/redo execution, level JSON serialization/deserialization, slot save/load persistence, statistics tracker event aggregation, settings save/load, and achievement unlocking conditions into a single 6-test suite. Updated `CMakeLists.txt` and verified 100% assertions pass.
- **Task 4: 2-Player & Shadow Mario Architecture Plan:** Authored detailed architectural design document `two_player_ai_plan.md` specifying Shared-Screen vs Split-Screen

viewports, AI Bot personalities (Speedrunner, Hunter, Collector), and Shadow Mario circular input buffer (`PlayerFramePacket` deque) for delayed ghost playback.

- **Task 5: Save/Load & Level Editor Integration Validation:** Ran interactive `SuperMarioGame` binary demo to verify real-time slot JSON generation (`slot_1.json` , `slot_2.json` , `slot_3.json`), settings persistence (`settings.json`), and seamless state switching between `PlayingState`, `MapEditor`, and `Pause HUD`.

25125084 - Trần Gia Huy (Member B - Input, Sound & Gameplay Systems)

- **Task 1: Input Keybindings Sync & Integration Verification:** Synchronized keybinding configuration structures between `InputManager` and the `Pause UI` settings panel. Tested input mapping stability during `dev` branch integration.

AI Usage Declaration

- Used AI to resolve `EventBus` subscriber handle sentinel logic during array mutation when unsubscribing listeners during event dispatches.
- Used AI to design a lightweight `verify_all.cpp` test harness combining unit test targets without external testing framework overhead.
- Used AI to design the circular packet buffer architecture (`PlayerFramePacket` deque) for Shadow Mario input recording and playback in `docs/two_player_ai_plan.md` .

Tasks Planned for Next Week

- **Nguyễn Đình Minh Huy:** Begin Phase 7 (Game States, Main Menu screen, Options menu UI, and World Map node path selection graph). Implement Time Rewind circular buffer memento state snapshots.
 - **Trần Gia Huy:** Implement Phase 10 In-Game Debug Overlay panel & Command Console (`~` key toggle), and assist in setting up 2-Player split-screen viewport rendering in `PlayingState` .
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Issues & Resolutions

Issue 1: EventBus Subscriber Listener Sentinel Bug on Unsubscribe

- **Problem:** When unsubscribing an EventBus listener handle in `PlayingState`, invalid handle checks using raw integer comparisons caused out-of-bounds array access or missed listener removals.
- **Resolution:** Updated `EventBus` subscriber handle sentinel representation to `static_cast<size_t>(-1)` to safely handle invalid or unsubscribed listener IDs across event broadcasts.

Issue 2: Stale Toast Rendering on Achievement Reset / Reload

- **Problem:** Loading a saved slot or invoking `setUnlockedAchievements()` did not clear active visual toast notifications in `AchievementManager`, causing previously unlocked toasts to pop up continuously after reloading a save.
- **Resolution:** Added explicit `m_activeToasts.clear()` inside `AchievementManager::setUnlockedAchievements()` and `reset()` to flush stale toasts prior to loading new slot states.

Issue 3: Save Slot Out-of-Bounds Flagpole and Spawn Point Coordinates

- **Problem:** Saving game state when player coordinates were out-of-bounds or near edge boundaries wrote raw negative or out-of-range floats into save slot JSON files, causing crash or invalid position loading.
- **Resolution:** Added `MathUtils::clamp` coordinate verification inside `Serializer.cpp` and `LevelLoader.cpp` to bound spawn points and flagpole coordinates strictly within level dimensions before JSON serialization.